

Initial Software Requirements Specification (SRS)

Student Accommodation Payment & Management System — Prototype

Group name:	Group C
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Stakeholder:	Gladys — owner/manager of 3 student accommodation properties
Date:	27/08/2026

1. Project Introduction

This project proposes an integrated prototype to help small student-accommodation providers manage rental payments, tenant information, maintenance requests and communication in one system, replacing manual spreadsheet and WhatsApp-based processes. The stakeholder for this project is Gladys Belle, who manages three student accommodation properties. The prototype consists of a Flutter mobile application for students, an ASP.NET web application for management, and a shared Supabase backend.

2. Stakeholder Profile and Current Process

2.1 Stakeholder Profile

Organisation	Graceful Plans
Organisation type	Small private student-accommodation provider (3 properties)
Stakeholder	Gladys Belle: Owner/manager
Main activities	Renting rooms to students, collecting rent, coordinating maintenance, communicating with tenants
Reason selected	Real, accessible stakeholder with a genuine manual-process problem across payments, tenant records and maintenance

2.2 Current Process

Gladys currently manages student accommodation information using an Excel spreadsheet. When a new student joins, their name, contact details, agreed rental amount and other relevant information are recorded after the contract is signed. Rental payments are manually recorded when received.

Day-to-day communication happens mainly through WhatsApp, since Gladys cannot regularly visit the properties in person. Students report problems and request assistance through WhatsApp. When maintenance is required, Gladys contacts a technician directly; physical site visits happen only occasionally.

2.3 Problems Experienced

- Tracking rental payments is the most time-consuming part of the process, taking approximately four hours and often done late at night.
- Students or their parents sometimes use incorrect or unclear payment references, making it difficult to identify who made a payment.
- Proof of payment must sometimes be requested manually, and students may take several days (or longer) to respond.
- Manual tracking can result in unpaid rent being missed, in one case, a student accumulated five months of unpaid rent because it was not followed up in time.
- Uncollected rent affects the business financially; Gladys has sometimes had to use her own salary to cover shortfalls.
- Maintenance requests sent privately through WhatsApp are easy to forget — both for Gladys and for the technician.
- Students have no way of knowing whether a maintenance problem they did not report has already been dealt with.
- There is no centralised way to see, at a glance, outstanding payments or outstanding maintenance tasks.

3. Problem Statement

Gladys currently uses an Excel spreadsheet and WhatsApp to manage rental payments, tenant records, and maintenance requests across three student accommodation properties. This results in payment tracking that takes approximately four hours per session, unpaid rent that can go unnoticed for months due to unclear payment references, and maintenance requests that are easily forgotten. The problem affects both Gladys and her student tenants because there is no organised, shared source of truth for payment status, payment history, or maintenance requests. An improved solution is needed to centralise rental payment tracking, provide clear payment references and history, and give both management and students visibility of maintenance requests and their status.

4. Project Aim and Objectives

Aim

To develop an integrated prototype that simplifies rental payment tracking, tenant record-keeping, and maintenance-request management for a small student-accommodation provider.

Objectives

- Allow students to view their payment reference, rental amount, rental period and payment history.
- Allow management to record and track rental payments against individual students.
- Provide management with a monthly report showing which students have paid and which have not.
- Allow students to submit maintenance requests and view the status of reported requests.
- Allow management to update the status of maintenance requests.
- Restrict access to payment and personal information to the relevant student and management only.

5. Project Scope

Deliverables	Flutter mobile prototype ASP.NET web prototype Supabase backend
Acceptance criteria	Payments recorded in ASP.NET are reflected in the student's Flutter view

	Maintenance requests submitted in Flutter appear in the ASP.NET dashboard and are visible to other students A monthly paid/unpaid report can be generated from recorded payment data
Exclusions	Real online payment processing / payment gateway integration (the system tracks payment status, not live transactions) Automated SMS/WhatsApp reminders (out of scope for the prototype)
Constraints	Must be completed within one semester Group's current technical skill level in Flutter, ASP.NET and Supabase Prototype must use test data rather than real student financial information
Assumptions	Gladys remains available for feedback and testing throughout the semester Three properties' worth of test data is sufficient to demonstrate the prototype

6. Stakeholder Needs

The following needs were identified and confirmed during the stakeholder interview conducted with Gladys (see Appendix B):

- Easier management of student accommodation and rental information.
- Clear, correct payment references for each student to reduce payment-matching errors.
- Rental payment records and a full payment history per student.
- A monthly report identifying which students have paid and which have not.
- A way to submit, view and track maintenance requests, including outstanding vs. completed status.
- Shared visibility of maintenance requests among students, so duplicate reports are avoided.
- Strict privacy of payment and personal information — visible only to the relevant student and management.
- OTP-based authentication rather than shared passwords.
- A system simple enough to use with limited technological experience, given appropriate training.

7. Functional Requirements

ID	Requirement	Type	Priority	Related stakeholder need
FR1	The student must be able to view their unique payment reference in the Flutter application.	Functional	Must Have	Correct payment references
FR2	The student must be able to view their rental amount, rental period and payment history.	Functional	Must Have	Payment history visibility
FR3	Management must be able to record a payment received against a specific student's account.	Functional	Must Have	Easier payment tracking
FR4	The system must associate each recorded payment with a student via their assigned payment reference.	Functional	Must Have	Difficulty matching payments
FR5	Management must be able to generate a monthly report showing which students have paid and which have not.	Functional	Must Have	Monthly payment reporting
FR6	The student must be able to submit a maintenance request describing the problem.	Functional	Must Have	Maintenance tracking

FR7	All students must be able to view submitted maintenance requests and their current status.	Functional	Must Have	Shared maintenance visibility
FR8	Management must be able to update the status of a maintenance request (outstanding / resolved).	Functional	Must Have	Outstanding task visibility
FR9	Management must be able to view a dashboard summarising outstanding payments and outstanding maintenance tasks.	Functional	Should Have	Centralised visibility
FR10	Management must be able to add and manage tenant records (name, contact details, room, rental period).	Functional	Must Have	Manual Excel record-keeping
FR11	Management must be able to post announcements visible to students.	Functional	Could Have	General communication needs

8. Non-Functional Requirements

ID	Requirement	Type	Priority	Related stakeholder need
NFR1	Users must authenticate using an OTP rather than a shared password.	Non-functional	Must Have	OTP over shared passwords
NFR2	Payment and personal information must be visible only to the relevant student and management, not to other students.	Non-functional	Must Have	Privacy of payment info
NFR3	The interface must be simple and understandable for a user with limited technological experience.	Non-functional	Must Have	Limited tech experience
NFR4	The prototype must use test data rather than real student financial or personal information.	Non-functional	Must Have	Responsible data handling
NFR5	Error messages must be clear and understandable to a non-technical user.	Non-functional	Should Have	Ease of use

9. System Requirements

ID	Requirement	Type	Priority	Related stakeholder need
SR1	The mobile application must be developed using Flutter.	System	Must Have	Module requirement
SR2	The web application must be developed using ASP.NET.	System	Must Have	Module requirement
SR3	The shared backend and database must use Supabase.	System	Must Have	Module requirement

10. Integration Requirements

ID	Requirement	Type	Priority	Related stakeholder need
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IR1	Payments recorded through the ASP.NET application must be stored in Supabase and reflected in the student's Flutter view.	Integration	Must Have	Shared payment status
IR2	Maintenance requests submitted through Flutter must be stored in Supabase and displayed in the ASP.NET dashboard.	Integration	Must Have	Shared maintenance tracking
IR3	Updates to maintenance-request status made in ASP.NET must be reflected in the Flutter application.	Integration	Must Have	Status visibility for students

11. Requirements Change Log

Date	Change	Reason
28/08/2026	Refined the payment requirement from a general "view rent status" function into an auto-matched payment reference and monthly paid/unpaid report.	Interview with Gladys identified unclear payment references and manual reference-matching as her single biggest pain point.
28/08/2026	Changed maintenance-request visibility from private (student-to-management only) to shared/general among students.	Gladys explained that shared visibility prevents duplicate reports and lets students see if an issue is already being handled.
28/08/2026	Selected OTP-based authentication instead of biometric login.	Biometric access is not available on every device; OTP was agreed as a workable alternative during the interview.

12. Refined Project Scope

Based on the stakeholder interview, the project scope taken forward into system design centres on three core areas: (1) rental payment recording, reference-matching and monthly reporting; (2) tenant record management; and (3) maintenance-request submission, shared visibility and status tracking all secured through OTP-based authentication and strict access control over payment information. Real online payment processing remains excluded; the system tracks payment status rather than handling live transactions.

Appendices

Appendix A — Stakeholder Interview Questions:

Stakeholder Interview Questions

Current Process

1. Can you explain how you currently manage students, rooms and accommodation information?
2. Can you walk us through the current process you follow when a student makes an accommodation payment, from the payment being made until it is recorded and confirmed?

Problems Experienced

3. Which parts of managing the accommodation are the most time-consuming or difficult for you, and why?
4. What problems, mistakes or delays tend to occur when using your current methods, and how do these affect your day-to-day work?

Intended Users

5. Who are the different people who would need to use an accommodation-management system, and what would each person use it for?
6. What information or tasks would you expect each type of user to access or manage?

Desired Mobile/Web Functions

7. Based on the challenges you have described, what tasks or processes would you most like to make easier, faster or more efficient?
8. Which tasks do students or staff currently perform manually or through platforms such as WhatsApp, Excel or email that you would prefer to manage in one place?

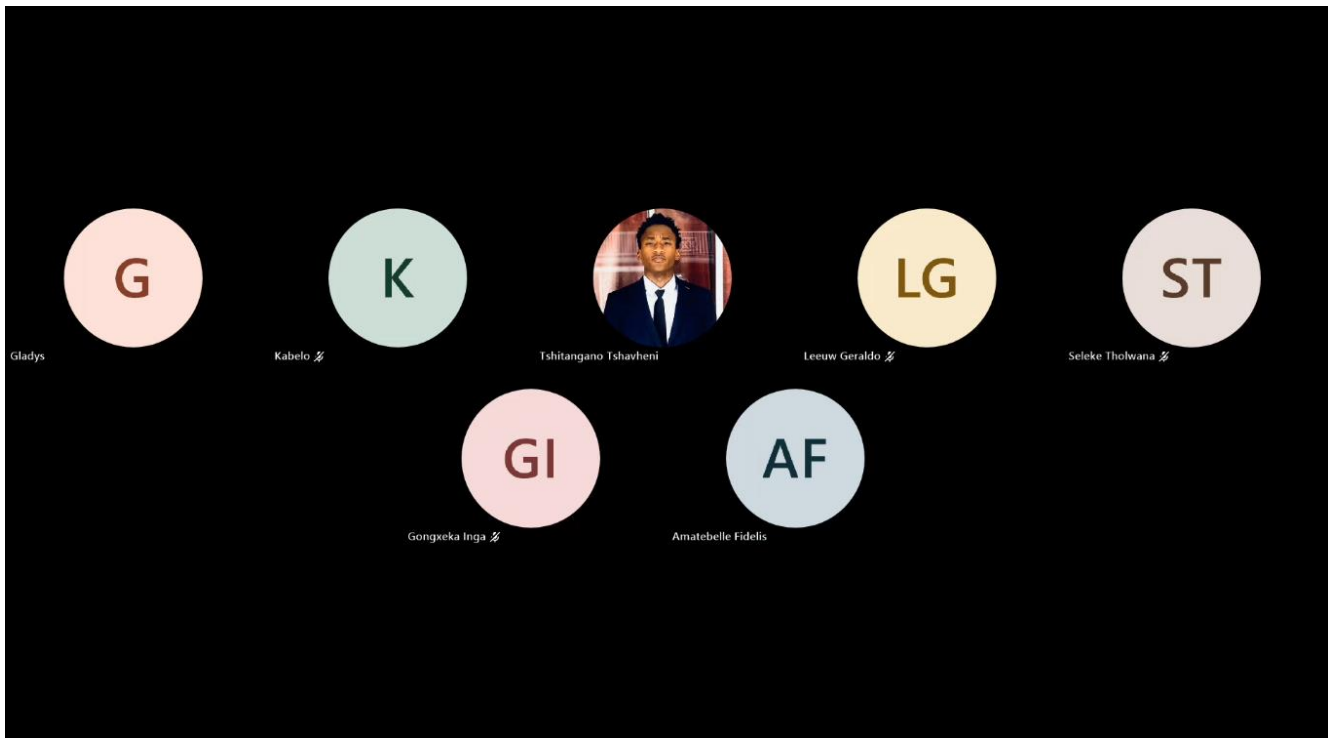
Information, Reports, Access and Security

9. What information or reports do you currently need to check regularly when managing the accommodation, and what would you like to be able to see more easily?
10. How should access to student, payment and accommodation information be managed, and what security or privacy requirements would you consider important?

Appendix B — Stakeholder Responses / Interview Notes:

□ [Interview notes pdf link](#)

Appendix C — Evidence of Stakeholder Engagement:



[1st Interview meeting with the Skakeholder 08000900-20260824 90255 AM-Meeting Recording.mp4](#)

Appendix D — Stakeholder Confirmation of Findings:

[PDF OF STAKEHOLDER FINDINGS](#)